

EST2000 - Digital Circuits

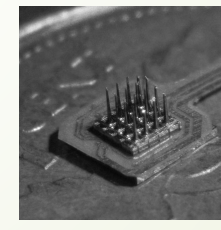
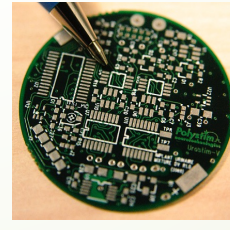
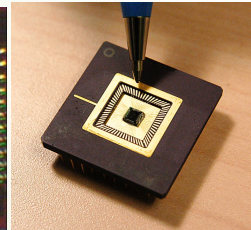
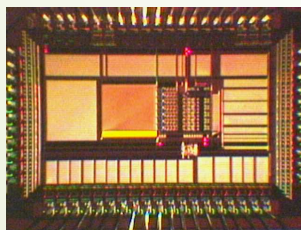
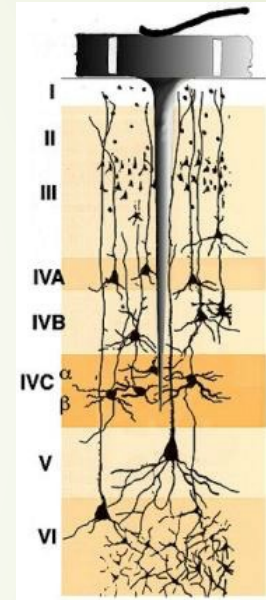
Fall 2025

Chapter 3

Week 2-3/16, Sept. 8th - 15th

(Agenda)

- Presentations
- Course (**Chapter 3**): Boolean algebra: theorems, and combinatorial circuits - II
- Other information.



1. Theory: Mondays, **E13-102**, 13:30-14:15; 14:20-15:05; 15:10-15:55
2. Lab works: Thursdays, **E10-301**, 13:30-15:05

Boolean algebra: theorems, and combinatorial circuits

Contents:

Chapter 2

1. Combinational Logic Outlines
2. George Boole, 1815 – 1864
3. Switching functions: Operators and Digital Logic Gates
4. Boolean Algebra Handy Tools
 - DeMorgan's Law, Consensus Theorem, ..

Chapter 3

5. Combinational Logic: Specifications Tools
6. Incompletely Specified Function

Reminder...

Consensus Theorem

• Boolean Algebra

$$\Rightarrow AB+AC+B'C$$

$$= AB+B'C$$

$$AB+AC+B'C$$

$$= AB+AC\mathbf{1}+B'C$$

$$= AB+AC(\mathbf{B+B'})+B'C$$

$$= AB+ABC+AB'C+B'C$$

$$= AB(\mathbf{1+C})+(\mathbf{A+1})B'C$$

$$= AB+B'C$$

$$\Rightarrow (A+B)(A+C)(B'+C)$$

$$= (A+B)(B'+C)$$

$$= (A+B)(A+C)(B'+C)$$

$$= (A+B)(A+C+BB')(B'+C)$$

$$= (A+B)(A+C+\mathbf{BB'})(B'+C)$$

$$= (A+B)(A+\mathbf{B})(B'+C)$$

$$= (A+B)(B'+C)$$

Reminder...

Consensus Theorem

Shannon's Expansion: Example

Which variable in $F(A,B,C)$ can be used for expansion ?

$$F(A,B,C) = AB' + AC + BC$$

- A. a
- B. b
- C. c
- D. None of the above

Review Summary: Switching Algebra & Karnaugh Map

Shannon's expansion and consensus theorem are used for logic optimization

- **Shannon's** expansion **divides** the problem into smaller functions
- **Consensus** theorem finds common terms when we **merge** small functions
- **Karnaugh** map mimics the above two operations in two-dimensional space as a visual aid.

5. Combinational Logic: Specifications

a. Language

b. Boolean Algebra

Canonical Expression: Sum of Minterms
and Product of Maxterms

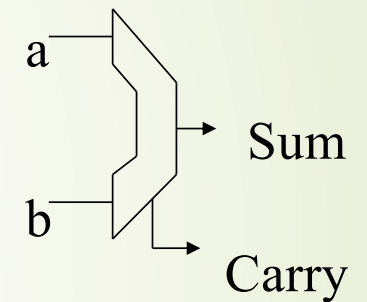
c. Truth Table: Minterms and Maxterms

d. Incompletely Specified Function.

5. Combinational Logic: Specifications (Cont'd)

- Decimal Addition

$$\begin{array}{r}
 5 \\
 + 7 \\
 \hline
 12 \\
 \text{Carry} \quad \text{Sum}
 \end{array}$$



- Binary Addition

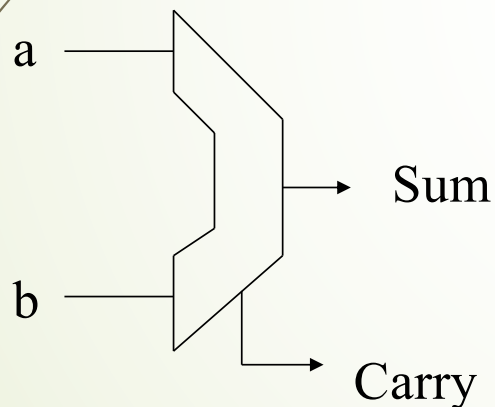
$$\begin{array}{r}
 1 \quad 1 \quad 1 \quad \text{Carry bits} \\
 1 \quad 0 \quad 1 \\
 + 1 \quad 1 \quad 1 \\
 \hline
 1 \quad 1 \quad 0 \quad 0 \\
 \text{Carry out} \quad \text{Sums}
 \end{array}$$

5
 7
 12

5. Combinational Logic: Specifications (Cont'd)

- Binary Addition: Hardware

➡ **Half Adder**: Two inputs (a,b) and two outputs (carry, sum).



Truth Table

id	a	b	carry	sum
0	0	0	0	0
1	0	1	0	1
2	1	0	0	1
3	1	1	1	0

5. Combinational Logic: Specifications (Cont'd)

- Binary Addition: Switching function

Switching Expressions:

$$\text{Sum } (a,b) = a' \cdot b + a \cdot b'$$

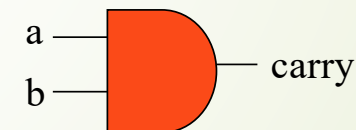
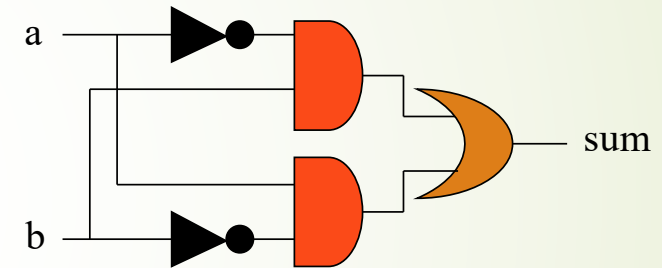
$$\text{Carry } (a, b) = a \cdot b$$

Ex:

$$\text{Sum } (0,0) = 0' \cdot 0 + 0 \cdot 0' = 0 + 0 = 0$$

$$\text{Sum } (0,1) = 0' \cdot 1 + 0 \cdot 1' = 1 + 0 = 1$$

$$\text{Sum } (1,1) = 1' \cdot 1 + 1 \cdot 1' = 0 + 0 = 0$$



5. Combinational Logic: Specifications (Cont'd)

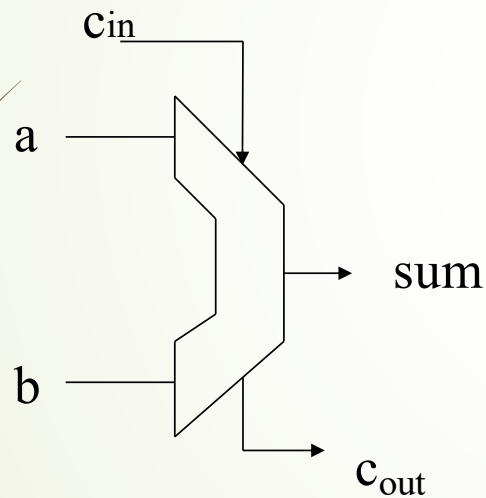
- Binary Addition: BSV notes

```
function Bit#(2) ha(Bit#(1) a, Bit#(1) b);  
  Bit#(1) s = (!a & b) | (a & !b);  
  Bit#(1) c = a & b;  
  return {c,s};  
endfunction
```

- ➡ {c, s} represents bit concatenation

5. Combinational Logic: Specifications (Cont'd)

- Full Adder: 3 inputs (a, b, c_{in}) and 2 outputs (carry, sum).



Arithmetic:
 $2c_{out} + \text{sum} = a + b + c_{in}$

id	a	b	c_{in}	Sum	c_{out}
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

5. Combinational Logic: Specifications (Cont'd)

- Minterms

A product of all variables in the function.

A minterm is equal to 1 on a row of the truth table.

id	a	b	c	Sum	C _{out}
0	0	0	0	0	0
1	0	0	1	$a'b'c$	0
2	0	1	0	$a'bc'$	0
3	0	1	1	0	$a'bc$
4	1	0	0	$ab'c'$	0
5	1	0	1	0	$ab'c$
6	1	1	0	0	abc'
7	1	1	1	abc	abc

5. Combinational Logic: Specifications (Cont'd)

- Maxterms

A sum of all variables in the function.

A maxterm is equal to 0 on a row of the truth table.

id	a	b	c	Sum	C _{out}
0	0	0	0	$a+b+c$	$a+b+c$
1	0	0	1	1	$a+b+c'$
2	0	1	0	1	$a+b'+c$
3	0	1	1	$a+b'+c'$	1
4	1	0	0	1	$a'+b+c$
5	1	0	1	$a'+b+c'$	1
6	1	1	0	$a'+b'+c$	1
7	1	1	1	1	1

5. Combinational Logic: Specifications (Cont'd)

Minterms and
Maxterms

Minterms cover all the outputs which are **true** (1).

Maxterms cover all the outputs which are **false** (0).

id	a	b	c	Minterms	Maxterms
0	0	0	0	$m_0=a'b'c'$	$M_0=a+b+c$
1	0	0	1	$m_1=a'b'c$	$M_1=a+b+c'$
2	0	1	0	$m_2=a'bc'$	$M_2=a+b'+c$
3	0	1	1	$m_3=a'bc$	$M_3=a+b'+c'$
4	1	0	0	$m_4=ab'c'$	$M_4=a'+b+c$
5	1	0	1	$m_5=ab'c$	$M_5=a'+b+c'$
6	1	1	0	$m_6=abc'$	$M_6=a'+b'+c$
7	1	1	1	$m_7=abc$	$M_7=a'+b'+c'$

5. Combinational Logic: Specifications (Cont'd)

- **Minterms**: Example

$$f_1(a,b,c) = a'bc + ab'c + abc' + abc$$

$$a'bc = 1 \text{ iff } (a,b,c) = (0,1,1)$$

$$ab'c = 1 \text{ iff } (a,b,c) = (1,0,1)$$

$$abc' = 1 \text{ iff } (a,b,c) = (1,1,0)$$

$$abc = 1 \text{ iff } (a,b,c) = (1,1,1)$$

$$f_1(a,b,c) = 1 \text{ iff } (a,b,c) = (0,1,1), (1,0,1), (1,1,0), \text{ or } (1,1,1)$$

$$\text{Ex: } f_1(1,0,1) = 1'01 + 10'1 + 101' + 101 = 1$$

$$f_1(1,0,0) = 1'00 + 10'0 + 100' + 100 = 0$$

5. Combinational Logic: Specifications (Cont'd)

- **Minterms:** $f_1(a,b,c) = a'bc + ab'c + abc' + abc$
- **Maxterms:** Example

$$f_2(a,b,c) = (a+b+c)(a+b+c')(a+b'+c)(a'+b+c)$$

$$a + b + c = 0 \text{ iff } (a,b,c) = (0,0,0)$$

$$a + b + c' = 0 \text{ iff } (a,b,c) = (0,0,1)$$

$$a + b' + c = 0 \text{ iff } (a,b,c) = (0,1,0)$$

$$a' + b + c = 0 \text{ iff } (a,b,c) = (1,0,0)$$

$$f_2(a,b,c) = 0 \text{ iff } (a,b,c) = (0,0,0), (0,0,1), (0,1,0), (1,0,0)$$

$$\text{Ex: } f_2(1,0,1) = (1+0+1)(1+0+1')(1+0'+1)(1'+0+1) = 1$$

$$f_2(0,1,0) = (0+1+0)(0+1+0')(0+1'+0)(0'+1+0) = 0$$

5. Combinational Logic: Specifications (Cont'd)

- Minterms and Maxterms (Cont'd)

$$f_1(a,b,c) = a'bc + ab'c + abc' + abc$$

$$f_2(a,b,c) = (a+b+c)(a+b+c')(a+b'+c)(a'+b+c)$$

$$f_1(a, b, c) = m_3 + m_5 + m_6 + m_7 = \text{Sm}(3,5,6,7)$$

$$f_2(a, b, c) = M_0M_1M_2M_4 = \text{PM}(0, 1, 2, 4)$$

Does $f_1 = f_2$?

$$f_1 = F1 = m_3 + m_5 + m_6 + m_7$$
$$f_1' = m_0 + m_1 + m_2 + m_4$$
$$f_1'' = m_0'.m_1'.m_2'.m_4' \Rightarrow f_1 = f_2 = M_0.M_1.M_2.M_4$$

The coverage of a single Minterm, e.g. $m_4 = ab'c'$

Id	a	b	c _{in}	Carry	Minterm ₄ = $ab'c'$	Maxterm ₄ = $a'+b+c$
0	0	0	0	0	0	1
1	0	0	1	0	0	1
2	0	1	0	0	0	1
3	0	1	1	1	0	1
4	1	0	0	0	1	0
5	1	0	1	1	0	1
6	1	1	0	1	0	1
7	1	1	1	1	0	1

Only one row has a 1.

Only one row has a 0.

5. Combinational Logic: Specifications (Cont'd)

- Minterms and Maxterms (Cont'd)

Summary

$$f_1(a,b,c) = a'bc + ab'c + abc' + abc$$

$$f_2(a,b,c) = (a+b+c)(a+b+c')(a+b'+c)(a'+b+c)$$

$$\text{Example : } f_1(a, b, c) = \sum \text{Minterms } (3,5,6,7)$$

$$f_2(a, b, c) = \prod \text{Maxterms } (0, 1, 2, 4)$$

Canonical representation of logic functions

Conversion between truth tables and switching functions

6. Incompletely Specified Function

Don't care set is important because it allows us to minimize the function

Id	a	b	f (a, b)
0	0	0	1
1	0	1	0
2	1	0	1
3	1	1	- (Don't care)

- 1) The input does not happen.
- 2) The input happens, but the output is ignored.

Examples:

- Decimal number 0... 9 uses 4 bits. The (1,1,1,1) does not happen.
- Final carry out bit (output is ignored).

6. Incompletely Specified Function (Cont'd)

$$\begin{aligned} g_1(a,b,c) &= a'b'c + a'bc + ab'c' + abc \\ &= m_1 + m_3 + m_4 + m_7 \\ &= \sum m(1,3,4,7) \end{aligned}$$

$$\begin{aligned} g_2(a,b,c) &= (a+b+c)(a'+b'+c) \\ &= M_0 M_6 \\ &= \prod M(0,6) \end{aligned}$$

id	a	b	c	$g_{1,2}$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	-
3	0	1	1	1
4	1	0	0	1
5	1	0	1	-
6	1	1	0	0
7	1	1	1	1

6. Incompletely Specified Function (Cont'd)

$$g_1(a,b,c) = \sum m(1,3,4,7)$$

$$g_2(a,b,c) = \prod M(0,6)$$

Does $g_1(a,b,c) = g_2(a,b,c)$?

$$g_1 = m_1 + m_3 + m_4 + m_7$$

$$g_1' = m_0 + m_2 + m_5 + m_6$$

$$g_1'' = m_0' \cdot m_2' \cdot m_5' \cdot m_6'$$

$$g_1 = M_0 \cdot M_2 \cdot M_5 \cdot M_6$$

id	a	b	c	$g_{1,2}$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	-
3	0	1	1	1
4	1	0	0	1
5	1	0	1	-
6	1	1	0	0
7	1	1	1	1

6. Incompletely Specified Function (Cont'd)

id	a	b	c	g_1	g_2
0	0	0	0	X	0
1	0	0	1	1	X
2	0	1	0	X	0
3	0	1	1	1	X
4	1	0	0	X	0
5	1	0	1	1	X
6	1	1	0	X	0
7	1	1	1	1	X

Does $g_1(a,b,c) = g_2(a,b,c)$?

$$g_1(a,b,c) = \sum m(1,3,5,7)$$

$$g_2(a,b,c) = \sum m(1,3,5,7) \text{ if } X=1$$

$$g_1(a,b,c) = m_1 + m_3 + m_5 + m_7$$

$$g_1'(a,b,c) = m_0 + m_2 + m_4 + m_6$$

$$g_1''(a,b,c) = m_0' m_2' m_4' m_6'$$

$$g_2(a,b,c) = \prod M(0,2,4,6)$$